

Impacts of Social Media on the Academic Performance of University Students in Pakistan

Niamat Ullah¹, Tehmoor Rehman², Sabiha Mengal², Amanullah³, Shafi Ullah⁴, Muhammad Mahroof Khan⁵ & Ahsan Ullah Kakar⁶

¹Research Assistant, Spatial Decision Support System (SDSS) Lab, National Center of GIS and Space Applications (NCGSA), Balochistan University of Information Technology, Engineering and Management Sciences (BUITEMS), Quetta 87300, Pakistan, Email: niamatullahza@gmail.com

²Lecturer, Department of Geography and Regional Planning, University of Balochistan, Quetta 87300, Pakistan, Email: sabihamengal1@gmail.com, tehmoor7@gmail.com

³M.Phil Scholar, Department of Statistics, University of Balochistan, Quetta 87300, Pakistan, Email: 96khaman@gmail.com

⁴Assistant Professor, Department of Computer Engineering, Balochistan University of Information Technology, Engineering and Management Sciences (BUITEMS), Quetta 87300, Pakistan, Email: shafi.ullah@buitms.edu.pk

⁵Lecturer, Department of Education, Alhamd Islamic University, Quetta 87300, Pakistan, Email: mhrfkhan@gmail.com

⁶Lecturer, Department of Education, University of Balochistan, Sub-Campus Pishin 87300, Pakistan, Email: ahsanullahkakar711@gmail.com

ARTICLE INFO

Article History:

Received:	February	10, 2025
Revised:	March	15, 2025
Accepted:	March	18, 2025
Available Online:	March	20, 2025

Keywords:

Social Media, Academic Performance, University Students, Balochistan, Pakistan

Corresponding Author:

Niamat Ullah

Email:

niamatullahza@gmail.com

ABSTRACT

Social media is a rapidly growing online category where users generate, share, bookmark, and network content. Social media has been a double-edged sword in education, and the introduction of smartphones has made it easy for students to use different platforms with ease. Though there is an overwhelming amount of evidence that points towards the facilitation of academic engagement through social networking by allowing peer-to-peer collaboration and sharing of resources overuse is associated with reduced academic productivity and heightened levels of stress. This research investigates the extent of social media implications in an academic setting, with specific emphasis on university students in Balochistan, Pakistan. Based on a quantitative method of a surveyed sample of 400 university students including undergraduate (72.3%), master (22.5%), and PhD (5.2%) from four major higher education institutions in Quetta, the provincial capital of Balochistan. The collected data was analyzed using Statistical Package for the Social Sciences (SPSS) and Microsoft Excel. The results of the present research identify trends in social media usage, time devoted, reasons for engagement, perceived academic impacts, and degree of distraction. Although a large number of respondents have seen social media as useful for academic purposes, distractions and the predominance of recreational use indicate a complicated relationship.



1. Introduction

Social media has become a vital tool in today's world, fostering peer connection and collaboration to promote learning and change (Asterhan & Bouton, 2017). The invention of the smartphone has increased the use of social media by enabling users to access many social networking sites with a few clicks (Kim & Khang 2014; Obi et al., 2012). Social media gives students a platform to share their everyday educational experiences, which improves academic performance (Tariq et al., 2012). However, social networking has negatively impacted students' academic performance (Obi et al., 2012; Rabin et al., 2011).

Excessive social media usage often harms academic productivity, this is the trend that has been observed in most studies (Goet, 2022). Addiction to social networking sites may result in higher levels of stress, which negatively impacts the student's educational achievements (Asanga et al., 2023; Azizi et al., 2019). Jamil et al. noted that spending too much time on social media results in lowering the time allocated for studying activities, which impacts learning achievement negatively (Jamil et al., 2020). Mensah and Nizam (2016) have shown that students who spend a lot of time on social media for informal chatting do poorly in their academic work (Mensah & Nizam, 2016). Kolhar et al., (2021) on the other hand, claim that students are more involved with gaming and entertainment on social media than with academic works, so they focused M and N's results and drew conclusions from other sources (Kolhar et al., 2021).

On the other hand, there are some studies that indicate social media has a positive effect on academic performance, especially when utilized for educational purposes (Aydoğmuş, 2023; D'Souza et al., 2017; Torun, 2020). For instance, Javeed (2022) established that although some students indicated adverse effects, others had positive impacts from interacting with educational content on sites such as Facebook and YouTube (Javeed, 2022). This dual nature is also seen in the research of Boahene et al., (2019) who theorize that social media has the ability to increase student participation and creativity, and therefore possibly lead to better academic performance (Boahene et al., 2019). In addition, Bukhari et al. highlight the need to account for individual characteristics like gender and proficiency, which may act as mediators of the relationship between academic performance and social media use (Bukhari et al., 2020).

In spite of all the available literature, there is still a significant research gap specifically in the context of Balochistan, Pakistan. Previous studies have been done in various educational and cultural contexts, which may not represent the specific socio-economic and educational problems of Balochistan students accurately. Moreover, although the literature addresses numerous factors affecting academic performance, there is not much research on how local cultural dynamics and education policies intersect with social media use. This is an area where future research can explore the particular effects of social media on academic performance among Balochistan university students, considering local variables and contexts.

2. Methodology

2.1 Study area

This research was carried out in Quetta city, which is the largest and capital city of province Balochistan. The city's population is 1565546 as per the 2023 census (PBS, 2024). Quetta is surrounded in the North by Pishin, in the East by Sibi in the South by Mustung and Chaghi districts, and the in West by Afghanistan through the Durand Line as shown in Figure 1. The average elevation of Quetta is about 1,680 meters (5,510 feet). Compared to the other areas of the province Quetta is more developed and has more education opportunities. The four largest

universities of Quetta were chosen for data collection regarding the effects of social media on the academic performance of students of universities. These institutions include the University of Balochistan (UoB), Balochistan University of Information Technology, Engineering and Management Sciences (BUITEMS), the Sardar Bahadur Khan Women University (SBKWU) and Alhamd Islamic University (AIU).

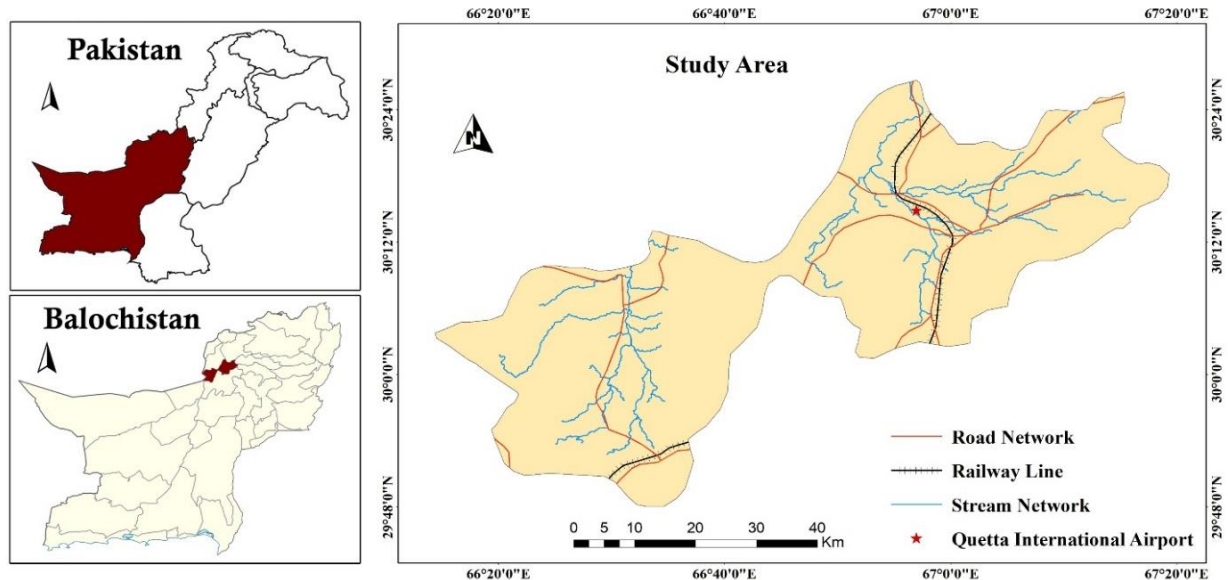


Figure 1: Location map of the study area.

The University of Balochistan, Quetta is the oldest higher education institution in the province. The University of Balochistan was set up in 1970. The University of Balochistan is the largest general university in the province providing higher education to the whole population of the province in science, arts, and humanities. Currently, the university consists of 06 faculties, 52 departments, 06 specialized centers, 57 affiliated colleges, 03 institutes, 04 campuses, and 01 constituent law college. BUITEMS establishment commenced in 2002, SBKWU began in 2004 whereas AIU was established in 2005.

2.2 Study design

This study employed the case study design to understand the impacts of social media on the academic achievement of university students.

2.3 Study population

Four prominent universities were chosen for the current study. The largest and the oldest university in the province is the University of Balochistan, Quetta with approximately 14000 enrolled students. It was founded in 1970. The second university chosen for the current study was Balochistan University of Information Technology, Engineering, and Management Sciences (BUITEMS) a public university in Quetta with 11000 students on its rolls and this university was founded in 2002. The third university chosen for the current study was Sardar Bahadur Khan Women's University (SBKWU) with more than 9000 students on its rolls and was established in 2004, the fourth and final university was Alhamd Islamic University (AIU) with total students of about 5000.

2.4 Sample design

the sample size of 400 students was chosen by employing the Yamane formula (Yamane, 1967). The given formula is as follows:

$$n = \frac{N}{1 + N(e)^2}$$

Where n = Sample size

N = total number of students

e = level of precision (5% = 0.05)

Sample = n = 40,000 = 399.99

$$1 + 40,000 (0.05)^2$$

The calculation from the Yamane formula gives 400 sample size from a population of 40,000. A stratified sampling method was adopted for collecting data from every stratum. A sample of 400 respondents was taken from three major universities in Quetta. The following formula of stratified sampling was adopted to obtain the population of each stratum.

$$\frac{\text{Sample size}}{\text{Entire population}} \times \text{Stratum size}$$

Based on the above formula we got the number of samples from each university shown in Table 1.

Table 1: Sampling size of the selected universities in the study area

University Name	Frequency	Percent
University of Balochistan (UoB), Quetta	140	35
Balochistan University of Information Technology Engineering and Management Sciences (BUIITEMS), Quetta	120	30
Sardar Bahadur Khan Women's University (SBKWU), Quetta	96	24
Alhamd Islamic University (AIU), Quetta	44	11
Total	400	100

2.5 Data collection and processing

Both primary and secondary sources of data were utilized in the current study. Primary data was gathered through the questionnaires from the respondents of selected universities. Together with this, secondary data were gathered from the selected universities regarding their total number of enrolled students and the boundaries of the district, province, and country. The primary data collected through questionnaires was further processed using Statistical Package for the Social Sciences (SPSS) and Microsoft Excel (MS Excel) for statistical analysis.

3. Results

3.1 Age of respondents

The age of respondents is shown in Figure 2 which presents the age distribution of respondents across four categories of age groups. The largest group (72%) falls within the second category (20-30 years), while 15% are below 20 years. The third group (30-40 years) accounts for 10%, and the smallest category represents just 3% of the total.

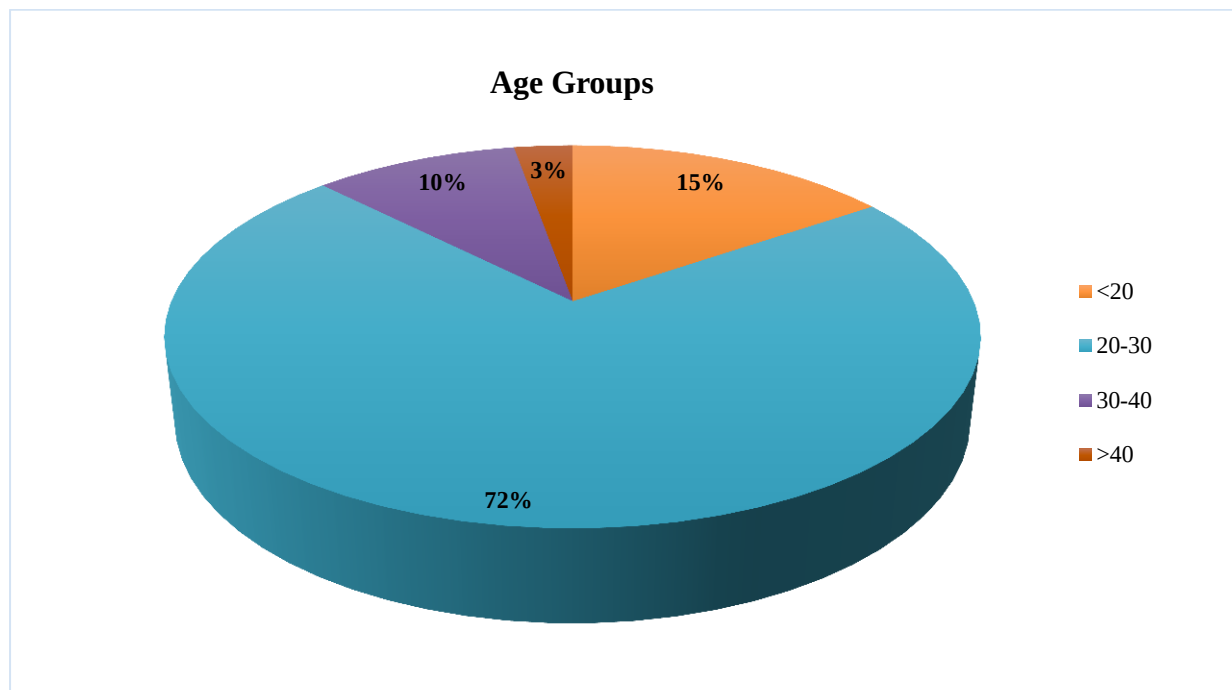


Figure 2: The age groups of the respondents

3.2 Gender distribution

The graph represents the gender breakdown in this research, with nearly equal numbers of male and female participants. This reflects a fair representation of both genders, providing equal participation in the study.

3.3 Educational level of respondents

The table showing the educational levels of the respondents shows that most (72.3%) have an undergraduate degree, making up 389 respondents. A lesser percentage, 22.5% (90 respondents), have a master's degree, while a very small percentage (5.2%) have achieved a PhD degree.

Table 2: The gender distribution and educational level of the respondents

Gender Distribution			Education Level		
Gender	Frequency	Percent	Education	Frequency	Percent
Male	198	49.5	Undergraduate	289	72.3
Female	202	50.5	Master	90	22.5
Total	400	100	PhD	21	5.2
			Total	400	100.0

3.4 Usage of social media platforms

The bar graph shows the respondents' use of social media, and their use spread across platforms. Facebook, TikTok, and Instagram are the leading social media platforms accounting for 18.6%, 16.1%, and 11.9% respectively. usage rate each at about 15-16%. Twitter (X) and Snapchat register a lower use rate of about 9.5% for both, while YouTube is at 12.8%. Interesting to note here is that large numbers of respondents (31.2%) use multiple platforms actively, thereby showing multi-platform use on a large scale.

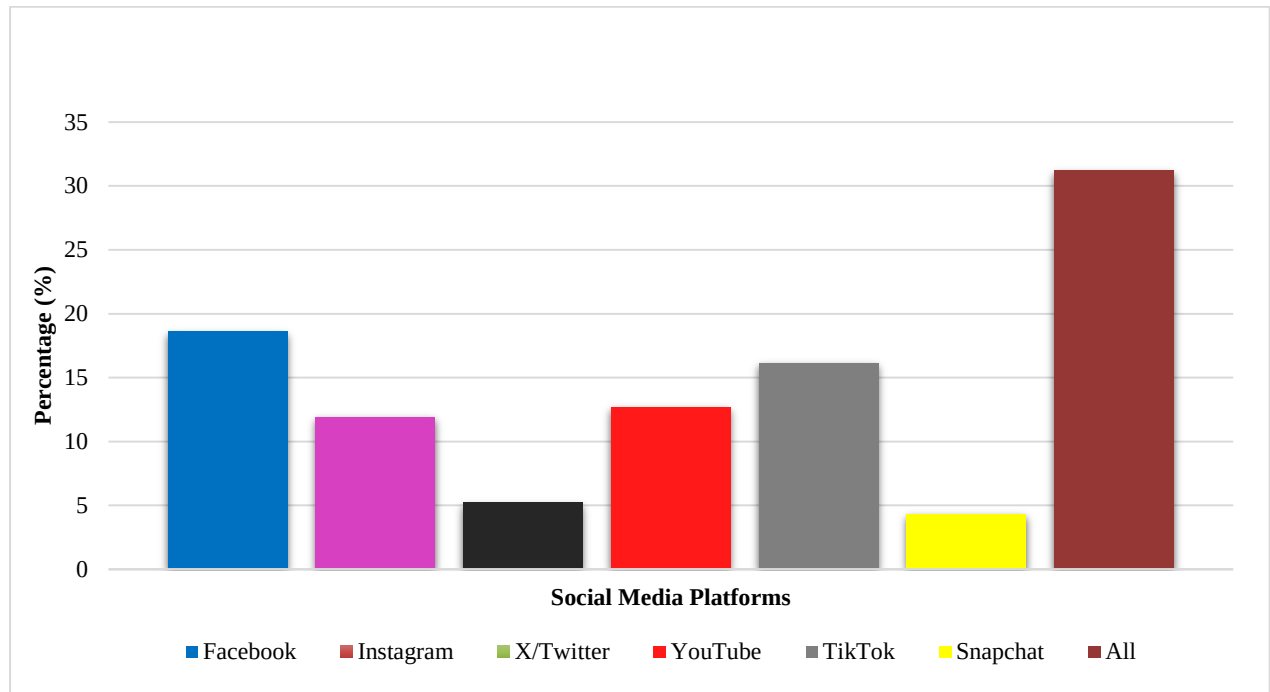


Figure 3: Usage of social media platforms

3.5 Average time spent on social media

The average time on social media represents the pattern of usage among the respondents on the different platforms. The results show that 19.6% of the respondents spend 3-4 hours a day on social media, which is the most prevalent duration. The second largest, with 33.6% of respondents, uses 2-3 hours per day. 27.6% of respondents have used social media for 1-2 hours per day. Moreover, 12% of the respondents spend over 5 hours, while 6.8% of the respondents spend less than one hour a day on social media as shown in figure 4.

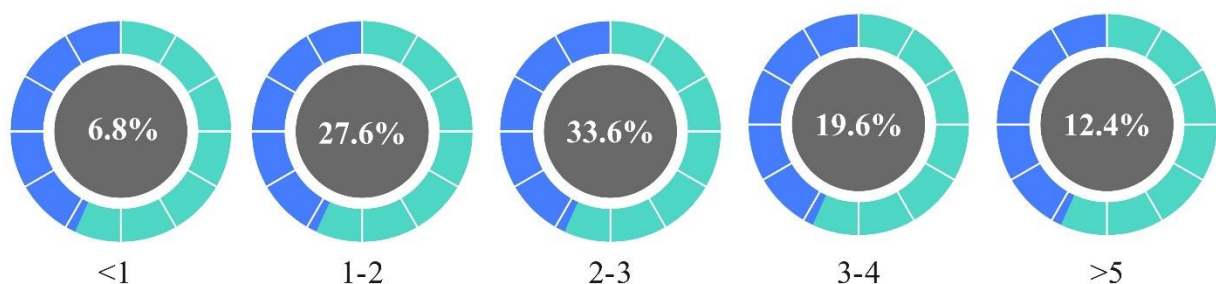


Figure 4. Average time spent on social media in one day.

3.6 Purpose of the social media

Table 3 shows the purpose for which social media is used by the respondents. The most frequent purpose, used by 39.7% (159 respondents), is new and entertainment. The second most frequent purpose is informational and academic use, used by 146 respondents (36.6%). The least frequent purpose is socializing with friends, used by 99 respondents (23.7%).

Table 3: The purpose of social media

Purpose	Frequency	Percent
Socializing with friends	99	23.7
Academic Purpose and Information	146	36.6
News and Entertainment	159	39.7
Total	400	100.0

3.7 Changes in academic performance

Figure 5 shows that the majority of the respondents (48.7%) have noticed a better performance in their studies by using social media sites. In contrast, 31.3% noticed a decrease in performance. Further, 12.5% indicated that social media did not affect their performance, and 7.5% were not sure, either because they were unaware or could not determine any changes.

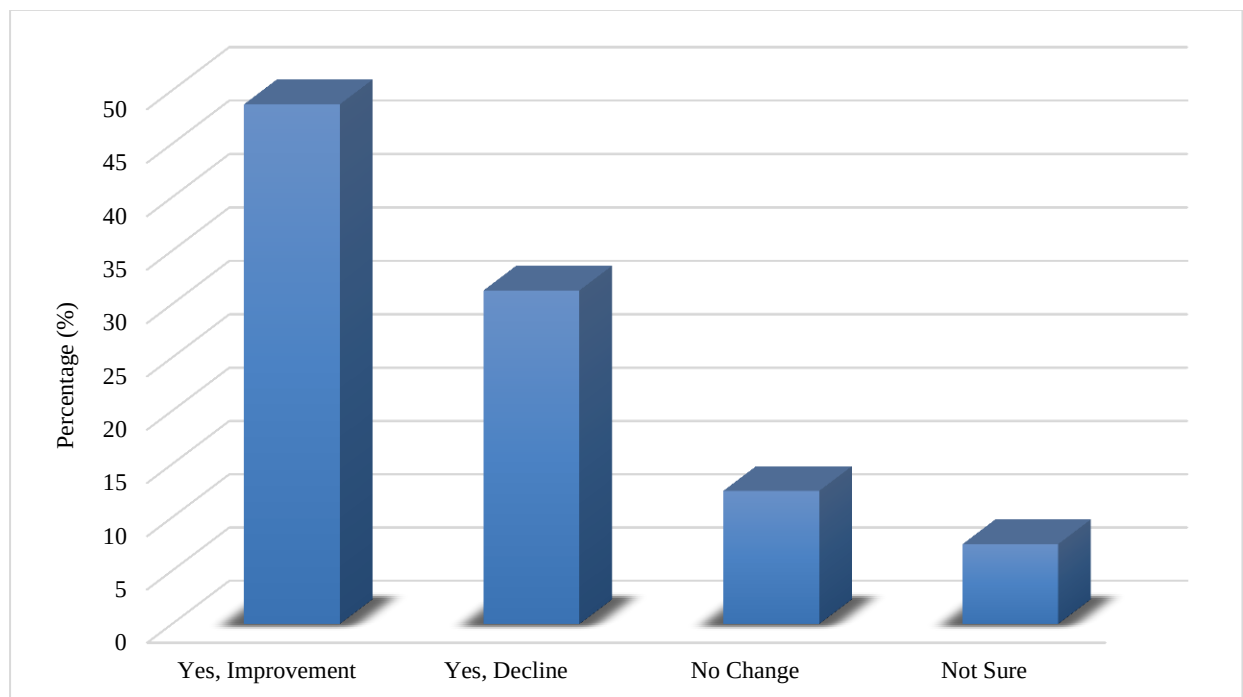


Figure 5: Change in academic performance of the respondents

3.8 Distracted by social media

Figure 6 shows statistics on the level to which the respondents are distracted by social media. Most, 42% (168 respondents), said they are sometimes distracted. In contrast, 17.1% (68 respondents) said they are rarely disturbed. A lesser percentage, 15.6%, confessed to always being distracted, while 13.6% (54 respondents) said they are never distracted by social media. Another 11.7% (47 respondents) said they are often distracted.

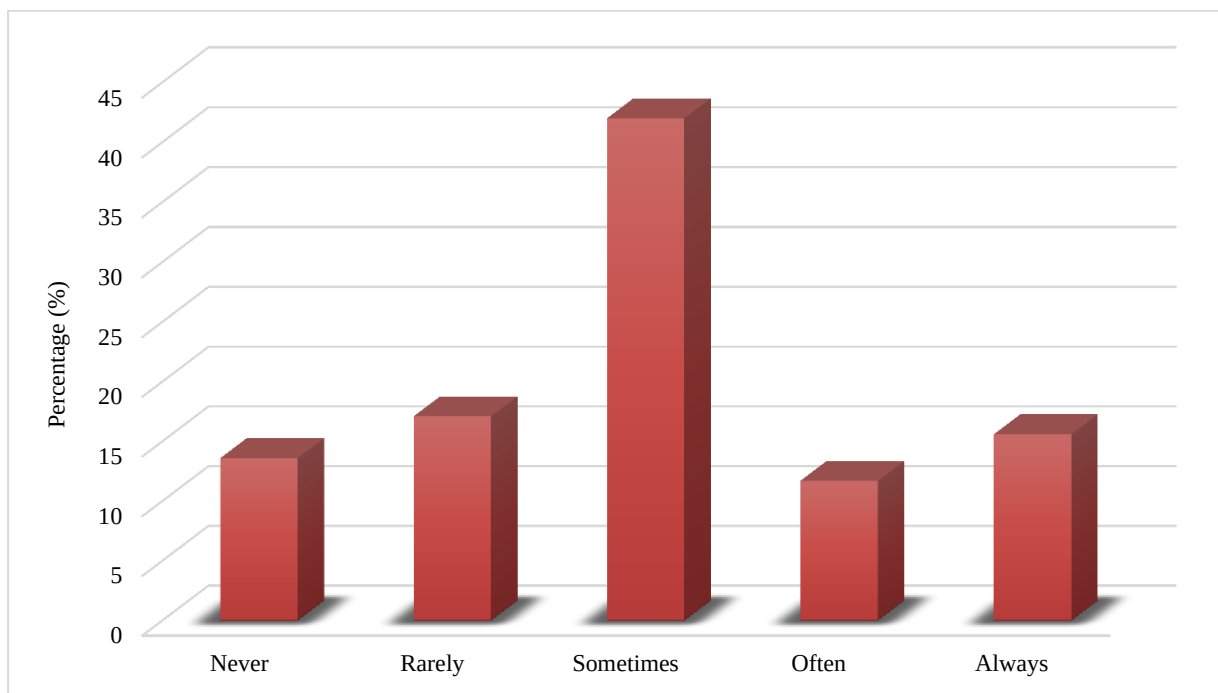


Figure 6: Distraction of the respondents by social media

3.9 Positive or negative impacts of social media

The figure 7 depicts student's views on the influence of social media on their performance. Close to half (45%) of the respondents hold the view that social media positively influences their studies. On the other hand, 32% view its influence as neutral in that they neither see any distinct advantages nor disadvantages. On the contrary, 14% perceive that social media adversely influences their performance, and they cite it as a distraction and a time-waster. Finally, 9% of the respondents are not sure if social media has affected their studies.

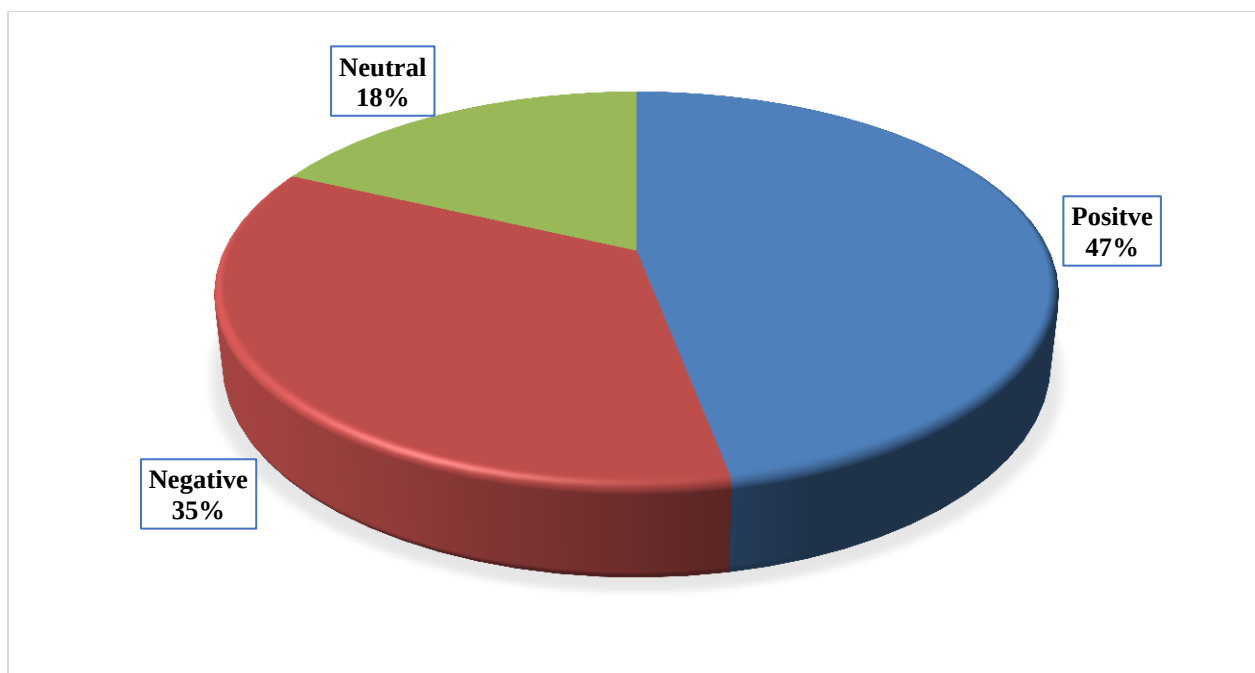


Figure 7: Impacts of social media on the students

4. Discussion

The results reveal that Facebook and Instagram are the most used platforms among students, with both of them being used by the respondents, while Twitter (X) and Snapchat show a much lower usage rate. This trend confirms previous research indicating that visual and interactive sites such as Facebook and Instagram are more attractive to young people, according to Katz and Nandi (2021) in the context of the COVID-19 pandemic in educational institutions (Katz & Nandi, 2021). Furthermore, the observed multi-platform usage in this study aligns with Gülbahar and Center (2014) findings about consolidating multiple social media platforms towards the enrichment of learning engagement to propose a more versatile way of learning (Gülbahar & Center, 2014).

Around 80% of participants indicated spending between 1 to 4 hours per day on social media. This is a confirmation of the research conducted by Alwagait et al., (2015) which revealed a large majority of students making use of social media within such time limits (Alwagait et al., 2015). The implications here point to the fact that while social media is an avenue for intellectual engagement, it also takes a large chunk of students' time, which might result in productive ends or diversions, according to Chen and Bryer (2012), who argue that social media use should be tightly controlled to bring about good learning environments (Chen & Bryer, 2012). The widespread application of social media for entertainment and news watching (39.7%) compared to educational use (36.6%) is significant. In contrast to evidence documenting how social media can aid academic collaboration, as presented by Boahene et al., (2019) in noting the positive correlation between academic performance and the use of social media, social media usage continues to be mostly recreational among students (Boahene et al., 2019). This implies that there needs to be more incorporation of learning content into such platforms to steer students' attention more towards educational engagement.

The finding of the study depicts that 47 of the participants believed that social media had a positive effect on their academic performance is an indication of increasing recognition of the usefulness of social media in achieving academic success through the sharing of resources and collaborative experiences. Caution has, however, been raised in literature against possible distractions by social media (Jamil et al., 2020). Research like Hrastinski and Aghaee (2011) described a "digital dissonance" among the perceptions of using social media by students for school versus personal reasons, which sustains this point (Hrastinski & Aghaee, 2011). The results reveal that 48.7% of the participants have occasional distractions from social media, which is in line with Nugraha and Dwita's (2023) findings, who recorded a strong relationship between social media multitasking and students' academic performance (Nugraha & Dwita, 2023). However, a smaller percentage reported they never get distracted, which highlights variation in the way people cope with their interaction with these platforms. The conflicting answers regarding distraction indicate the need for an organized approach wherein educational institutes can direct students to acquire improved time management skills (Torun, 2020).

Lastly, close to half of the respondents see social media as a positive factor in their studies, which indicates an optimistic attitude towards its impact in educational settings. This is in accordance with what (Manca & Ranieri, 2016) hypothesize, that although social media is not wholesome in all cases for learning purposes, there is significant potential among students to use these media well. The acknowledgment of social media as a double-edged sword with the potential for benefits and distractions urges a more sophisticated understanding of how educational institutions can enable its effective use.

4.1 Recommendations

The study has provided some recommendations that are totally based on the findings that are explained in the previous chapter. The recommendations suggested by this study are as follows

- Teachers should make sure that students use social media to enhance their learning capabilities.
- The time students spend on using the social media should be reduced to be more focused on the academics.
- Policy formation to control and manage the use of social media in university to not lose interest in their studies.

5. Conclusion

Social media is a powerful learning tool, its effectiveness is highly dependent on how and where it is used. This study was conducted in the top four universities of Balochistan to assess the impacts of social media on the academic performance of the students. Primary data was collected from university students including undergraduate, master and PhD through questionnaire method and the data was further processed in SPSS and MS Excel for statistical analysis. All variables used in the present study were constructed, and also the data was entered through binary coding as 0, 1, 2, etc. Likewise, data results were shown in the form of frequencies and percentages. The results of the present research confirm both the positive and negative effects of the use of social media on the academic performance of students, with a significant proportion of the respondents claiming that social media improves their learning experience. However, the problems introduced by distraction and recreational use cannot be ignored. The inconsistencies highlighted in the manner in which students engage with social media emphasize the need for learning institutions to craft formal strategies aimed at fostering constructive interaction with such sites, blending recreational and pedagogical content.

6. Policy Implication and Future Research

This study adds to the existing debate surrounding the use of social media in education by offering proof of its widespread use by students, its functions, and its complexities regarding learning performance. Future research must prioritize devising means to counter the distractions offered by social media while tapping its learning potential, as suggested by the subtle information from existing and previous scholarly works.

Acknowledgments: The authors are grateful to the Spatial Data Support System (SDSS) Lab, NCGSA, BUITEMS, for providing guidance and expert support for the present study.

Author Contribution: Niamat Ullah, Tehmoor Rehman, Sabiha Mengal; wrote, designed, analyzed, and interpreted the data; Amanullah, Shafi Ullah, Mahroof Khan, Ahsanullah; contributed materials, analysis tools, or data; wrote the paper.

Funding: The current research has not received any funding.

References

1. Alwagait, E., Shahzad, B., & Alim, S. (2015). Impact of social media usage on students academic performance in Saudi Arabia. *Computers in Human Behavior*, 51, 1092-1097. <https://doi.org/10.1016/j.chb.2014.09.028>

2. Asanga, M. P., Essiet, U. U., Ukhurebor, K. E., Afolorunso, A., & Hussaini, P. (2023). Social media and academic performance: a survey research of senior secondary school students in Uyo, Nigeria. *International Journal of Learning, Teaching and Educational Research*, 22(2), 323-337. <https://doi.org/10.26803/ijlter.22.2.18>
3. Asterhan, C. S., & Bouton, E. (2017). Teenage peer-to-peer knowledge sharing through social network sites in secondary schools. *Computers & Education*, 110, 16-34.
4. Aydoğmuş, M., Tut, E., & Karadağ, Y. (2023). Teachers' Experiences Regarding the Use of Social Media for Educational Purposes. *International Journal of Psychology and Educational Studies*, 10(1), 69–82. <https://doi.org/10.52380/ijpes.2023.10.1.855>
5. Azizi, S.M., Soroush, A. & Khatony, A. (2019). The relationship between social networking addiction and academic performance in Iranian students of medical sciences: a cross-sectional study. *BMC Psychology* 7(28). <https://doi.org/10.1186/s40359-019-0305-0>
6. Boahene, K., Fang, J., & Sampong, F. (2019). Social media usage and tertiary students' academic performance: examining the influences of academic self-efficacy and innovation characteristics. *Sustainability*, 11(8), 2431. <https://doi.org/10.3390/su11082431>
7. Bukhari, M., Mahmud, R., & Manan, N. (2020). Social media: does usage have impact on academic performance? *Universal Journal of Educational Research*, 8(10), 4416-4420. <https://doi.org/10.13189/ujer.2020.081008>
8. Chen, B. & Bryer, T. (2012). Investigating instructional strategies for using social media in formal and informal learning. *The International Review of Research in Open and Distributed Learning*, 13(1), 87. <https://doi.org/10.19173/irrodl.v13i1.1027>
9. D'Souza K, Henningham L, Zou R, Huang J, O'Sullivan E, Last J, Ho K. (2017). Attitudes of Health Professional Educators Toward the Use of Social Media as a Teaching Tool: Global Cross-Sectional Study. *JMIR Medical Education*; 3(2):e13. doi: 10.2196/mededu.6429.
10. Goet, J. (2022). Impact of social media on academic performance of students. *KIC International Journal of Social Science and Management*, 1(1), 35-42.
11. Gülbahar, Y., & Center, D. E. (2014). Current state of usage of social media for education: Case of Turkey. *Journal of Social Media Studies*, 1(1), 53-69
12. Hrastinski, S. & Aghaee, N. (2011). How are campus students using social media to support their studies? an explorative interview study. *Education and Information Technologies*, 17(4), 451-464. <https://doi.org/10.1007/s10639-011-9169-5>
13. Jamil, M., Ain, Q. ul, Batool, S., Saadat, S., Malik, S., Arshad, M., Nagra, R. N., Haider, M., Shameem, R., & Latif, B. (2020). Impact of Social Media on Academic Performance. *European Journal of Medical and Health Sciences*, 2(5). <https://doi.org/10.24018/ejmed.2020.2.5.512>
14. Javeed, A. (2023). Impact of Using Social Media on the Academic Performance: A Case of College Going Students in Loralai Division. *Siazga Research Journal*, 2(1), 88–95. <https://doi.org/10.58341/srj.v2i1.12>
15. Katz, M., & Nandi, N. (2021). Social media and medical education in the context of the COVID-19 pandemic: scoping review. *JMIR Medical Education*, 7(2), e25892. doi: 10.2196/25892.
16. Kim, Y., & Khang, H. (2014). Revisiting civic voluntarism predictors of college students' political participation in the context of social media. *Computers in Human Behavior*, 36, 114-121
17. Kolhar, M., Kazi, R, N, A. & Alameen, A. (2021). Effect of social media use on learning, social interactions, and sleep duration among university students. *Saudi Journal of Biological Sciences*, 28(4), 2216-2222. <https://doi.org/10.1016/j.sjbs.2021.01.010>.

18. Manca, S. and Ranieri, M. (2016). "Yes for sharing, no for teaching!": social media in academic practices. *The Internet and Higher Education*, 29, 63-74. <https://doi.org/10.1016/j.iheduc.2015.12.004>
19. Menseh, SO. and Nizam, S. (2016) The Impact of Social Media on Students' Academic Performance: A Case of Malaysia Tertiary Institution. *International Journal of Education, Learning and Training*, 1, 14-21. <https://doi.org/10.24924/ijelt/2016.11/v1.iss1/14.21>
20. Nugraha, M. & Dwita, V. (2023). The influence of social media usage on loyalty with customer satisfaction & brand trust as mediation variable. *Banking & Management Review*, 11(2), 1616-1640. <https://doi.org/10.52250/bmr.v11i2.541>
21. Obi, N. C., Bulus, L. D., Adamu, G. M., & Sala'at, A. B. (2012). The need for safety consciousness among youths on social networking sites. *Journal of Applied Science and Management (JASM)*, 14(1), 29.
22. Pakistan Bureau of Statistics. (2023). *Population & Housing Census 2023*. Government of Pakistan
23. Rabin, L. A., Fogel, J., & Nutter-Upham, K. E. (2011). Academic procrastination in college students: The role of self-reported executive function. *Journal of clinical and experimental neuropsychology*, 33(3), 344-357.
24. Tariq, W., Mehboob, M., Khan, M. A., & Ullah, F. (2012). The impact of social media and social networks on education and students of Pakistan. *International Journal of Computer Science Issues (IJCSI)*, 9(4), 407.
25. Torun, E. (2020). Educational use of social media in higher education: gender and social networking sites as the predictors of consuming, creating, and sharing content. *Acta Educationis Generalis*, 10(2), 112-132. <https://doi.org/10.2478/atd-2020-0013>
26. Yamane, T. (1967). *Statistics: An Introductory Analysis* (2nd ed.). Harper and Row, New York.